ML-DL-OPS Lab Assignment 1 Report

Important Links Regarding the Assignment -

Google Colab Notebook(.ipynb) - https://colab.research.google.com/drive/1wy1002SXKjy-abjaqT9PWI_MTTV3OrTS?usp=sharing
GitHub Repository - https://github.com/Layaa-V/MLOps-LayaaVishwakarma-M25CSA017/tree/Assignment-1

Q1(a) - Analysis of Deep Learning Models -

The following table summarizes the classification test accuracy
for ResNet-18 and ResNet-50 across various hyperparameters, including Batch Size, Optimizer, and Learning Rate.

Batch Size	Optimizer	Learning Rate	ResNet-18 Accuracy (%)	ResNet-50 Accuracy (%)
16	SGD	0.001	98.87	98.59
16	SGD	0.0001	95.27	89.31
16	Adam	0.001	99.39	99.18
16	Adam	0.0001	99.13	99.00
32	SGD	0.001	98.34	97.98
32	SGD	0.0001	92.51	67.95
32	Adam	0.001	99.36	99.07
32	Adam	0.0001	99.10	97.54

Impact of Optimizers and Learning Rates -

- **Adam vs. SGD**: The results indicate that Adam is highly stable across different learning rates, consistently achieving accuracy above 99%.
- **SGD Sensitivity**: SGD showed a significant performance degradation when the learning rate was reduced to 0.0001, particularly with the deeper ResNet-50 model at a batch size of 32 (Accuracy: 67.95%).
- **Learning Rate (LR)**: A learning rate of 0.001 was optimal for both models, while 0.0001 often led to slower convergence or poor local minima for SGD-based training.

Model Complexity and Generalization -

- **ResNet-18:** This architecture provided the best balance between speed and accuracy for simple digit classification, consistently meeting 80%+ accuracy.
- **ResNet-50:** Although more powerful, ResNet-50 required more precise hyperparameter tuning to avoid the "vanishing gradient" or convergence issues observed at lower learning rates.

Q1(b) - Analysis of Classical ML Models (SVM) -

This section reports the performance of a Support Vector Machine (SVM) classifier using Polynomial and RBF kernels on MNIST and FashionMNIST datasets. The following table summarizes the findings.

Dataset	Kernel	Test Accuracy (%)	Train Time (ms)
MNIST	poly	96.41	42719.35
MNIST	rbf	96.45	60662.33
FashionMNIST	poly	86.19	47858.95
FashionMNIST	rbf	85.16	58355.74

- **Observations on MNIST Dataset**: On the MNIST dataset, the RBF and Polynomial kernels performed similarly (approx. 96%), but the Polynomial kernel was significantly faster to train, as seen from the table.
- **Observations on FashionMNIST Dataset**: Accuracy dropped to around 85-86% on FashionMNIST, highlighting that clothing items possess more complex features than handwritten digits, making them harder for SVMs to classify.
- **Kernel**: Across both datasets, the Polynomial kernel was consistently faster to train than the RBF kernel while maintaining comparable accuracy.

Q2 - Comparative Hardware and Computational Efficiency -

The following table summarizes the findings of Q2 -

Compute	Batch Size	Optimizer	Learning Rate	Model	Test Accuracy (%)	Train Time (ms)	FLOPs
GPU	16	SGD	0.001	ResNet-18	87.64	2239.423513	142441984.0
GPU	16	Adam	0.001	ResNet-18	88.65	2287.928343	142441984.0
GPU	16	SGD	0.001	ResNet-50	80.92	4807.822227	330881024.0
GPU	16	Adam	0.001	ResNet-50	84.77	4548.972845	330881024.0
CPU	16	SGD	0.001	ResNet-18	87.77	9764.726877	142441984.0
CPU	16	Adam	0.001	ResNet-18	89.48	11360.593557	142441984.0
CPU	16	SGD	0.001	ResNet-50	78.84	18040.843010	330881024.0
CPU	16	Adam	0.001	ResNet-50	74.02	21231.782675	330881024.0

CPU vs. GPU Efficiency -

- **Performance Gap**: The experiment shows a massive disparity in training time between the CPU and GPU. For ResNet-50, the training time was reduced from 21,231 ms (CPU) to ~4,548 ms (GPU). This represents a speedup.
- **Parallelism**: This difference is attributed to the architectural design of the GPU, which contains thousands of cores optimized for simultaneous matrix multiplications, whereas the CPU is designed for serial task processing.
- **Accuracy Consistency**: Despite the change in compute device, the test accuracy remained relatively consistent (e.g., 87.64% on GPU vs. 87.77% on CPU for ResNet-18). This confirms that hardware affects speed (efficiency) but does not inherently alter the mathematical capability of the model.

Optimizer Performance on Different Hardware -

- **Adam's Advantage**: On both CPU and GPU, the Adam optimizer generally produced higher accuracy than SGD within the same timeframe.
- **Overhead**: While Adam is more computationally efficient in terms of convergence, it slightly increased the training time per batch (e.g., 11,360 ms for Adam vs. 9,764 ms for SGD on CPU) because it maintains additional moving averages for gradients.

(The visualization graphs for train-val loss and accuracy are included in the collar notebook).